EXPONENTIAL and LOGARITHMIC FUNCTIONS

LECTURE

by

JK GONDWE

LECTURE OBJECTIVES

- Solving exponential and logarithmic equations
- Identify graphs of exponential growth and decay

Techniques used

Techniques based on whether the equations have the variable in the *exponent* (exponential equation) or inside a *logarithm* (logarithmic equation) :

EXPONENTIAL EQUATIONS

2 terms with different bases

- place one term on either side of the equation.
- Change the terms to same base then use the property :

if $a^x = a^y$ then $x = y$

Note: there is only one base and power on each side and the bases are the same.

.

Example

Solve $4^x = 8^{\frac{4}{3}}$

$$(2^2)^x = (2^3)^{\frac{4}{3}}$$

$$2^{2x} = 2^4$$

$$\therefore \quad 2x = 4$$

$$x = 2$$

Cont'd

If the 2 terms with different bases, on either side the equation cannot be changed to have the same base then either

- take the log of both sides and solve or
- write the exponential equation in log form and solve

Cont'd

$$2^x = 7$$

$$\rightarrow \log 2^x = \log 7$$

or

$$\rightarrow \log_2 7 = x \quad (\text{Note}: \text{use change of base to solve})$$

Logarithmic equations

Equation with variable inside the log:

- if there are 2 terms with the same base and NO OTHER TERMS, get one log on each side of the equation then apply the property;
- if $\log_a x = \log_a y$ then $x=y$

  Since both sides of the equation are equal, the logarithms are equal, hence what is inside each log must be equal.

Cont'd

- If there is another term besides the logs, combine any log terms and get them on one side of the equation.
- Put the other term on the other side.
- Write that logarithmic equation in EXPONENTIAL form and solve.

Cont'd

$$\log_2 x + \log_2 (x\text{-}7) = 3$$

$$\log_2 (x)(x-7) = 3$$

$$2^3 = (x)(x-7)$$

$$x = 8 \; or \; -1$$

Example

Solve $5^{x-1} = 12$

Recall

If $\ln x = \ln y$, then $x = y$.

Similarly, if $5^{x-1}=12$

then $\log 5^{x-1}=\log 12$

cont'd

$$\log 5^{x-1} = \log 12$$

$$(x-1)\log 5 = \log 12$$

$$x-1 = \frac{\log 12}{\log 5}$$

$$x = \frac{\log 12}{\log 5} + 1$$

$$x = 2.54$$

Solve $\log_3(2x+5)=2.$

Rewrite in exponential form:

$$3^2 = 2x+5$$
$$9 = 2x+5$$
$$4 = 2x$$
$$\boxed{2 = x}$$

Exponential growth and decay

- These are rates which represent the change in some quantity through time
- Exponential growth - increase in number or size over time
- Exponential decay - decrease in number or size over time

Exponential growth

- Function in the form $f(x) = ab^x$, with a>0 and b>1
  - The function increases as *x* increases

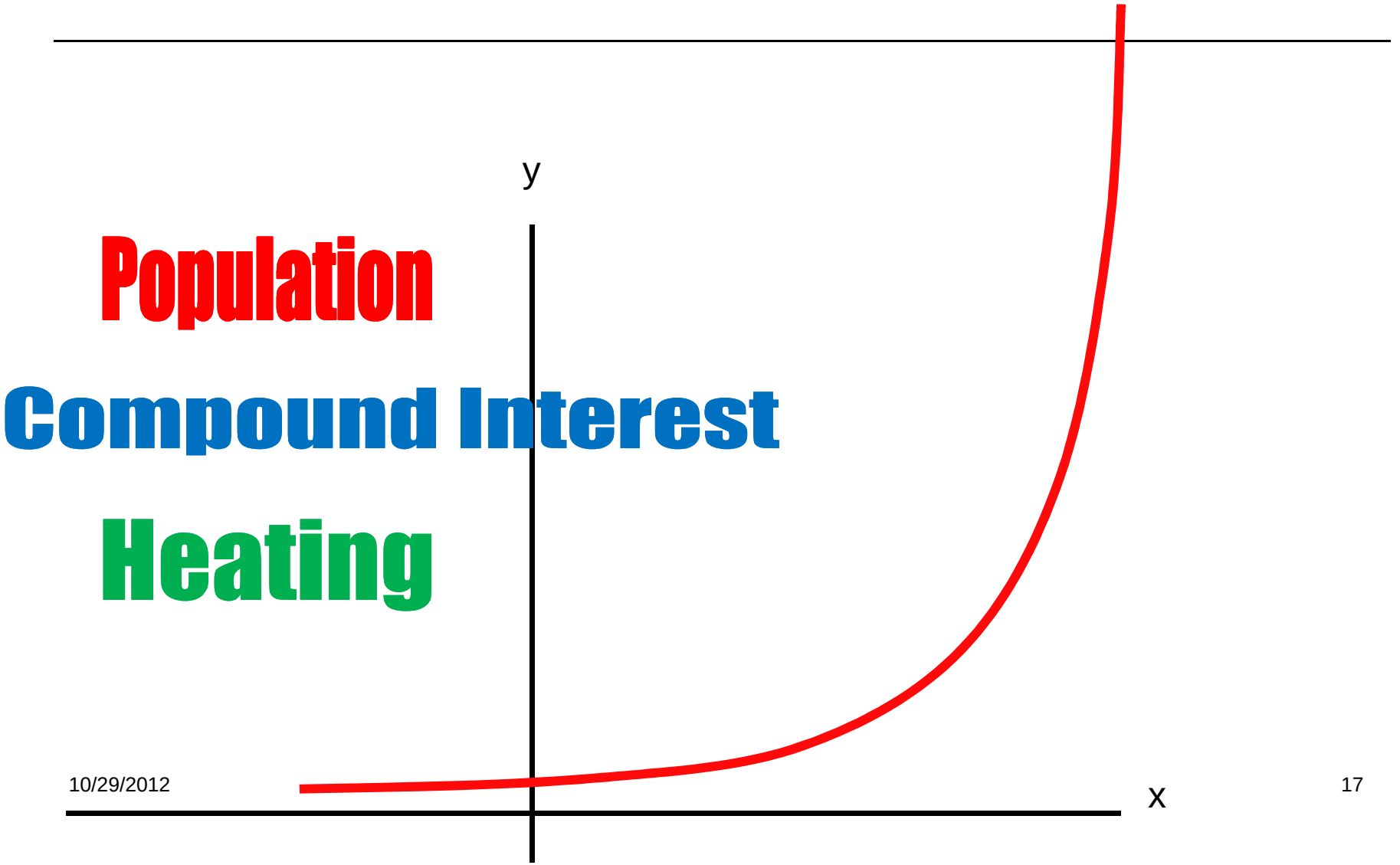
    eg the growth of bacteria colonies.

cont'd

If N_o is the initial population and the growth rate is k then the population N at time t is:

$N = N_o e^{kt}$ where $k > 0$

Graph of exponential growth



Exponential decay

- Function in the form $f(x) = ab^x$, with a>0 and b>1
  - The function decreases as *x* increases

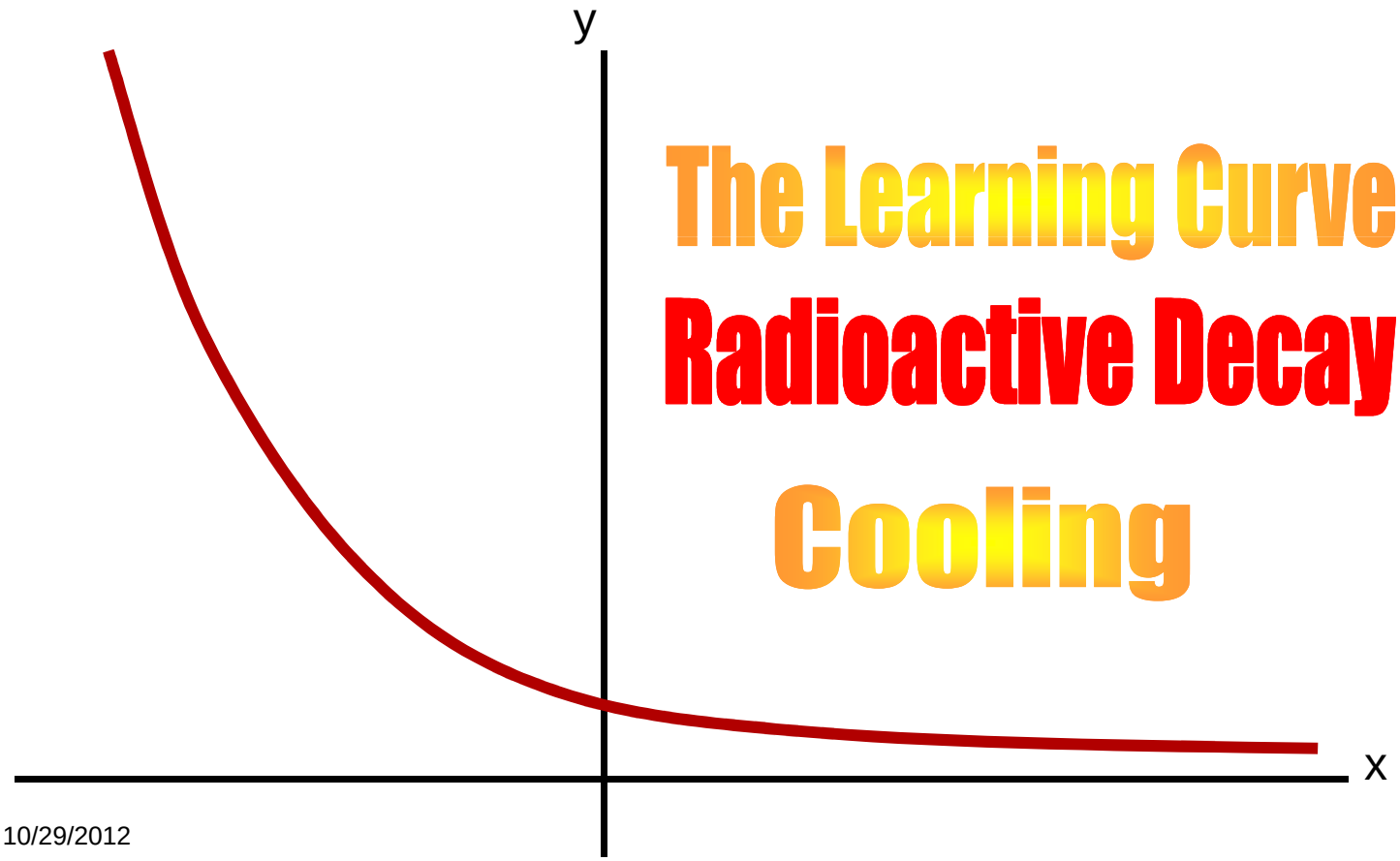
    eg radioactive decay

Exponential Decay

If N_o is the initial population and the decay rate is k then the population N at time t is:

$N = N_o e^{kt}$ where $k<0$

Graph of Exponential Decay



Exercise

1. Solve the simultaneous equations $\log(x-2)+\log 2 = 2\log y$

   $\log(x-3y+3) = 0$

2. Solve for x, correct answer to two significant figures,

   the equation $4^x - 2^{x+1} - 3 = 0$

3. Suppose the population of a colony of bacteria increases exponetially. At the start of an experiment, there are 6,000 bacteria, and one hour later, the population has increased to 6,400. How long will it take for the popution to reach 10,000? Round your answer to the nearest hour.